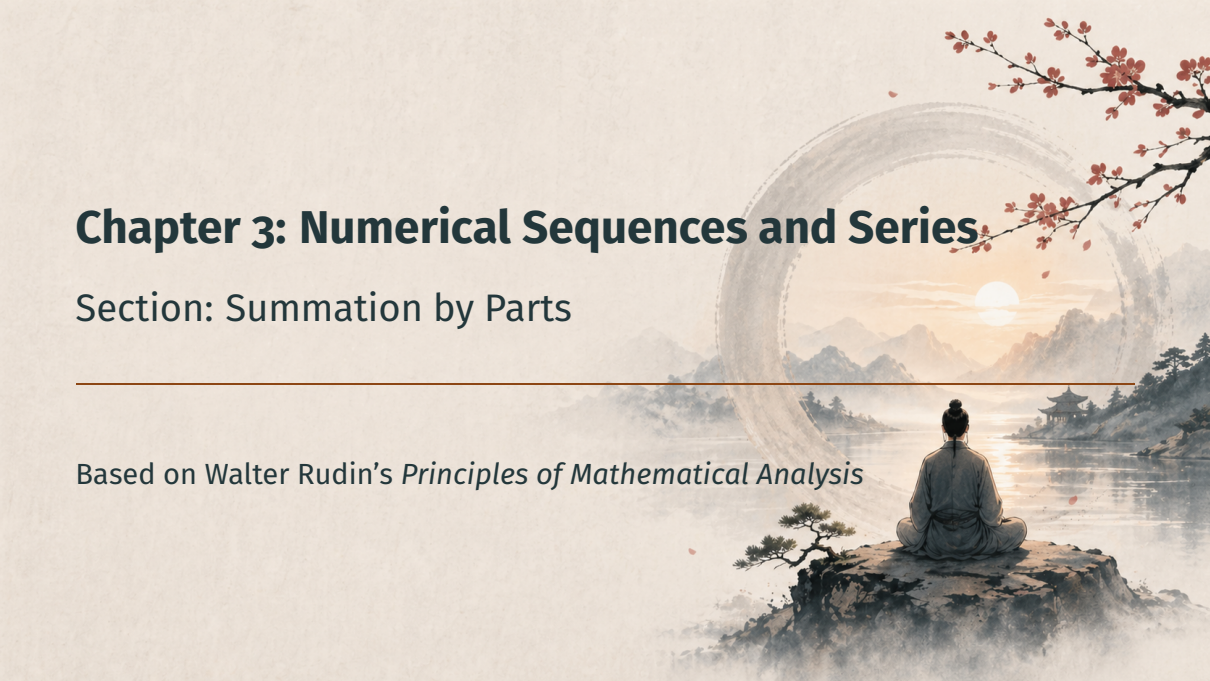


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person with their hair in a bun, wearing a grey robe, sits in a meditative lotus position on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains are visible. A large, faint circular frame, resembling a yin-yang symbol, is centered in the background. A small pavilion with a curved roof sits on the far shore. Red cherry blossom branches with small flowers hang down from the top right corner. A thin horizontal line is positioned below the section header.

Section: Summation by Parts

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

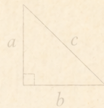
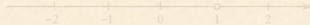
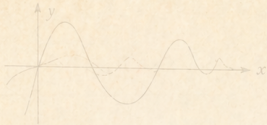
The Partial Summation Formula

Dirichlet's Test

Alternating Series

Power Series on the Boundary

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Summation by Parts?

- Goal: understand series $\sum a_n b_n$ — a sum of **products**



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why Summation by Parts?

- Goal: understand series $\sum a_n b_n$ — a sum of **products**
- Continuous analogue: **integration by parts**

$$\int u \, dv = uv - \int v \, du$$

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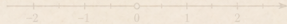
$$\int u \, dv = uv - \int v \, du$$

- Payoff: convergence tests when $\{b_n\}$ is **monotonic** (Dirichlet, alternating series, boundary of power series)

$$f(x) = \sin x$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The Partial Summation Formula



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Theorem 3.41: Partial Summation Formula

Theorem 3.41

Given sequences $\{a_n\}$, $\{b_n\}$, set

$$A_n = \sum_{k=0}^n a_k \quad (n \geq 0), \quad A_{-1} = 0.$$

Then for $0 \leq p \leq q$,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \quad (20)$$

Proof of 3.41 — Step 1: Express a_n via partial sums

Step 1: Telescoping the coefficients

Since $A_n = \sum_{k=0}^n a_k$ and $A_{-1} = 0$,

$$a_n = A_n - A_{n-1}.$$

Therefore

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^q (A_n - A_{n-1}) b_n.$$

Proof of 3.41 — Step 2: Split and reindex

Step 2: Two sums, shifted index

Split the sum:

$$\sum_{n=p}^q (A_n - A_{n-1})b_n = \sum_{n=p}^q A_n b_n - \sum_{n=p}^q A_{n-1} b_n.$$

Reindex the second sum ($m = n - 1$):

$$\sum_{n=p}^q A_{n-1} b_n = \sum_{n=p-1}^{q-1} A_n b_{n+1}.$$

Proof of 3.41 — Step 3: Collect the terms

Step 3: Match the overlapping range

$$\sum_{n=p}^q A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1}$$

Overlap on $p \leq n \leq q-1$ gives $A_n(b_n - b_{n+1})$; the leftover endpoints give $A_q b_q$ and $-A_{p-1} b_p$:

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

Theorem 3.41: A Useful Identity

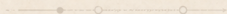
Remark

Formula (20) is the **partial summation formula** (Abel summation).

It is most useful for $\sum a_n b_n$ when $\{b_n\}$ is **monotonic**: then the differences $b_n - b_{n+1}$ all have the **same sign**.



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Dirichlet's Test



Theorem 3.42: Dirichlet's Test

Theorem 3.42

Suppose

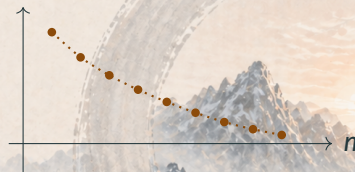
- (a) the partial sums A_n of $\sum a_n$ form a **bounded** sequence;
- (b) $b_0 \geq b_1 \geq b_2 \geq \dots$ (monotone decreasing);
- (c) $\lim_{n \rightarrow \infty} b_n = 0$.

Then $\sum a_n b_n$ converges.

Theorem 3.42: Reading the Hypotheses



$|A_n| \leq M$
(bounded)



$b_n \searrow 0$
(monotone, $\rightarrow 0$)

Proof Strategy: Theorem 3.42

- **Tool:** the partial summation formula (20).
- **Goal:** verify the **Cauchy criterion**: make $\left| \sum_{n=p}^q a_n b_n \right|$ small for large p .
- **Key fact:** boundedness of A_n plus monotonicity of b_n collapses the estimate to a **multiple of b_p** .

Proof of 3.42 — Step 1: Apply formula (20) and bound

Step 1: Insert the partial summation formula

Choose M with $|A_n| \leq M$ for all n . For $N \leq p \leq q$:

$$\begin{aligned} \left| \sum_{n=p}^q a_n b_n \right| &= \left| \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p \right| \\ &\leq M \left(\sum_{n=p}^{q-1} (b_n - b_{n+1}) + b_q + b_p \right). \end{aligned}$$

Proof of 3.42 — Step 2: Telescope and conclude

Step 2: The sum telescopes

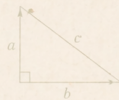
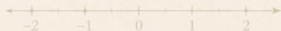
The differences telescope: $\sum_{n=p}^{q-1} (b_n - b_{n+1}) = b_p - b_q$. Hence

$$\left| \sum_{n=p}^q a_n b_n \right| \leq M(b_p - b_q + b_q + b_p) = 2M b_p.$$

Given $\varepsilon > 0$, pick N with $b_N \leq \varepsilon/(2M)$. Then for $p \geq N$,

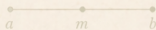
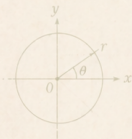
$$\left| \sum_{n=p}^q a_n b_n \right| \leq 2M b_p \leq 2M b_N \leq \varepsilon.$$

By the Cauchy criterion, $\sum a_n b_n$ **converges**.



$$a^2 + b^2 = c^2$$

Alternating Series



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.43: Alternating Series Test

Theorem 3.43 (Leibniz)

Suppose

- (a) $|c_1| \geq |c_2| \geq |c_3| \geq \cdots$;
- (b) $c_{2m-1} \geq 0, c_{2m} \leq 0 \quad (m = 1, 2, 3, \dots)$;
- (c) $\lim_{n \rightarrow \infty} c_n = 0$.

Then $\sum c_n$ converges.

Terminology

Series satisfying (b) — signs $+, -, +, -, \dots$ — are **alternating series**.

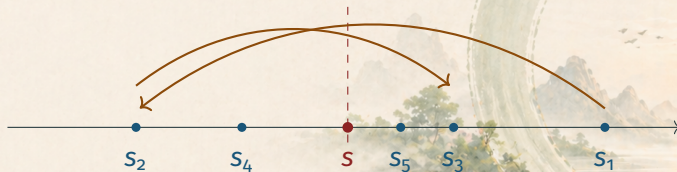
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Alternating Series: Why It Converges

odd s_n decrease, even s_n increase — nesting onto s



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

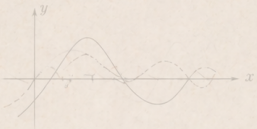
Proof of 3.43 — A Corollary of Dirichlet

Proof: apply Theorem 3.42

Set $a_n = (-1)^{n+1}$ and $b_n = |c_n|$. Then $c_n = a_n b_n$.

- Partial sums of a_n : $A_n \in \{0, 1\}$ — **bounded**. ✓ (a)
- $b_n = |c_n|$ is **decreasing**. ✓ (b)
- $b_n = |c_n| \rightarrow 0$. ✓ (c)

Hypotheses of 3.42 hold, so $\sum c_n = \sum a_n b_n$ converges. ■

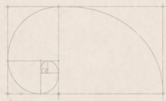


$$e^{i\theta} = \cos \theta + i \sin \theta$$



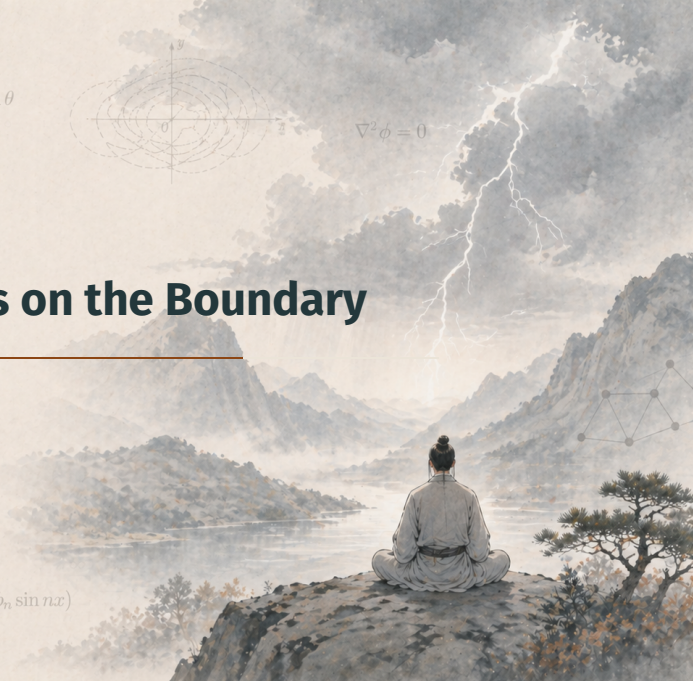
$$\nabla^2 \phi = 0$$

Power Series on the Boundary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



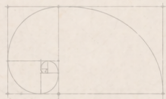
Theorem 3.44: Convergence on the Unit Circle

Theorem 3.44

Suppose $\sum c_n z^n$ has radius of convergence 1, and

$$c_0 \geq c_1 \geq c_2 \geq \cdots, \quad \lim_{n \rightarrow \infty} c_n = 0.$$

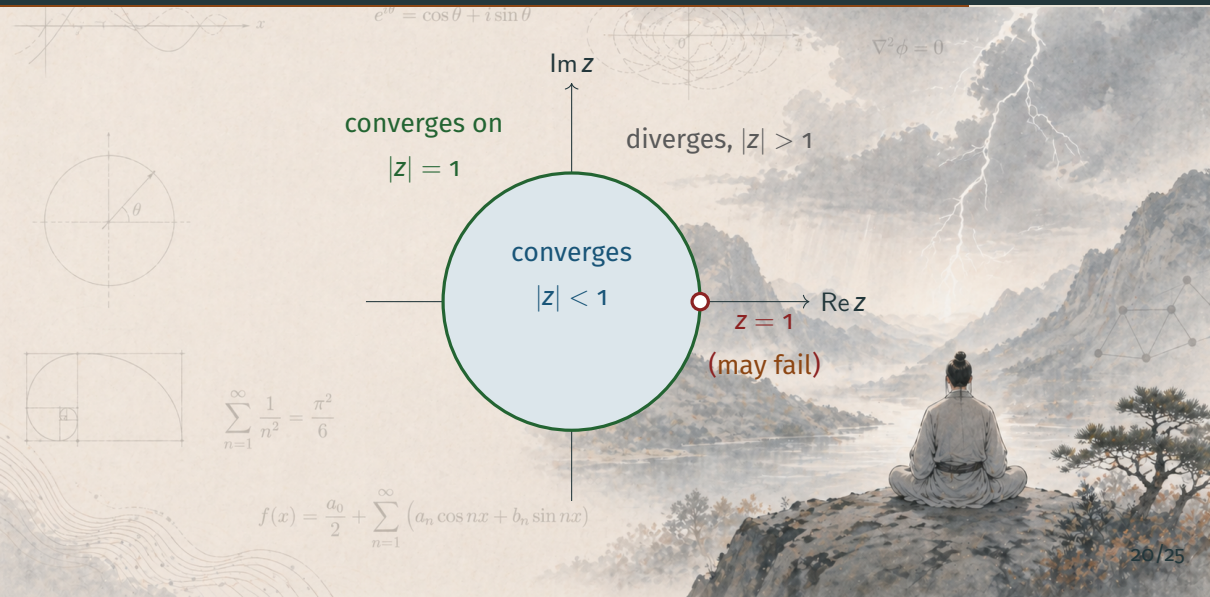
Then $\sum c_n z^n$ converges at **every** point on the circle $|z| = 1$, **except possibly** at $z = 1$.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Theorem 3.44: The Picture on the Unit Circle



Proof of 3.44 — Bounding the Geometric Partial Sums

Proof: apply Theorem 3.42 with $a_n = z^n$

Put $a_n = z^n$, $b_n = c_n$. For $|z| = 1$, $z \neq 1$:

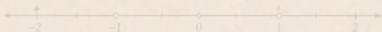
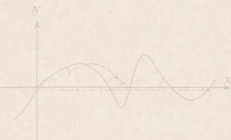
$$|A_n| = \left| \sum_{m=0}^n z^m \right| = \left| \frac{1 - z^{n+1}}{1 - z} \right| \leq \frac{2}{|1 - z|}.$$

So $\{A_n\}$ is **bounded**, while $b_n = c_n$ decreases to 0.

By Theorem 3.42, $\sum c_n z^n$ converges. ■

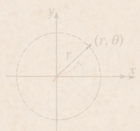
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$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Summary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Partial summation (3.41):** the discrete integration by parts,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

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Summary

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- **Partial summation (3.41):** the discrete integration by parts.
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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



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- **Partial summation (3.41):** the discrete integration by parts.
- **Dirichlet's test (3.42):** if A_n bounded and $b_n \searrow 0$, then $\sum a_n b_n$ converges.
- **Alternating series (3.43):** decreasing $|c_n| \rightarrow 0$ with alternating signs $\Rightarrow \sum c_n$ converges.
- **Power series (3.44):** $c_n \searrow 0 \Rightarrow \sum c_n z^n$ converges on $|z| = 1$, except possibly at $z = 1$.

Coming up next: Absolute Convergence.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

